

Chapter 09: Phase Diagrams & The Iron-Carbon System

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Materials Science (MIAE 221)

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapter 9

1. Executive Overview & First-Principles Philosophy

In materials engineering, the macroscopic mechanical properties of an alloy—its strength, hardness, toughness, and ductility—are not governed solely by chemical composition. They depend overwhelmingly on its **microstructure**: the number of phases present, their relative amounts, and their spatial distribution (morphology).

A **Phase Diagram** (also called an **equilibrium diagram**) is the graphical "road map" of metallurgy:

- It maps the thermodynamically stable phases that exist at any combination of **temperature, pressure, and chemical composition** under equilibrium conditions.
- It enables metallurgical engineers to predict:
 1. What phases are present at temperature T .
 2. The exact chemical composition of each phase.
 3. The precise weight fraction of each phase using the **Inverse Lever Rule**.
 4. The microstructural evolution during solidification and heat treatment.

In ferrous metallurgy, the **Iron-Carbon ($\text{Fe} - \text{Fe}_3\text{C}$) Phase Diagram** forms the scientific bedrock of all carbon steels and cast irons—the structural backbone of civil infrastructure, transportation, and industrial machinery.

2. Fundamental Definitions & Phase Equilibria (Callister §9.2 – §9.5)

2.1 Component vs. Phase vs. Microstructure

- **Component:** Pure chemical elements or chemically stoichiometric compounds that constitute an alloy system (e.g., in a brass alloy, the components are Cu and Zn ; in carbon steel, the components are Fe and C).
- **Phase:** A physically distinct, chemically homogeneous, and mechanically separable portion of matter:
 - Pure water containing ice cubes is a **two-phase system** (solid ice + liquid water), even though it has only **one component** (H_2O).
 - A solid solution of Ni dissolved in Cu is a **single-phase solid** (α), even though it has **two components**.
- **Solubility Limit:** The maximum concentration of solute atoms that can dissolve into a solvent matrix at a specific temperature without forming a new second phase (analogous to the saturation limit of sugar dissolving in tea).

2.2 Gibbs Phase Rule (Callister §9.15)

The **Gibbs Phase Rule** defines the number of degrees of freedom (F)—the number of independent intensive variables (temperature, pressure, composition) that can be varied simultaneously without altering the number of phases in equilibrium:

$$P + F = C + 2$$

In metallurgical systems, pressure is held constant at atmospheric pressure ($P_{\text{atm}} = 1 \text{ atm}$), eliminating one degree of freedom. This yields the **Condensed Gibbs Phase Rule**:

$$P + F = C + 1$$

where:

- P = number of phases present in thermodynamic equilibrium.
- F = degrees of freedom.
- C = number of chemical components.
 - In a single-phase region ($P = 1, C = 2$): $F = 2 - 1 + 1 = 2$ (Both temperature and composition can vary independently).
 - In a two-phase region ($P = 2, C = 2$): $F = 2 - 2 + 1 = 1$ (Specifying temperature automatically fixes the compositions of both phases).
 - At an invariant reaction point ($P = 3, C = 2$): $F = 2 - 3 + 1 = 0$ (Zero degrees of freedom: reaction occurs at one unique temperature and composition).

3. Binary Isomorphous Systems: The Inverse Lever Rule (Callister §9.7 – §9.9)

A binary system is **isomorphous** when the two components exhibit complete, unlimited liquid and solid solubility across all compositions from 0 to 100 (e.g., the Copper-Nickel Cu-Ni system).

with binary alloys—those that contain two components. If more than two components are present, phase diagrams become extremely complicated and difficult to represent. An explanation of the principles governing and the interpretation of phase diagrams can be demonstrated using binary alloys even though most alloys contain more than two components.

Binary phase diagrams are maps that represent the relationships between temperature and the compositions and quantities of phases at equilibrium, which influence the microstructure of an alloy. Many microstructures develop from *phase transformations*, the changes that occur when the temperature is altered (typically upon cooling). This may involve the transition from one phase to another or the appearance or disappearance of a phase. Binary phase diagrams are helpful in predicting phase transformations and the resulting microstructures, which may have equilibrium or nonequilibrium character.

ISOMORPHOUS SYSTEMS

Possibly the easiest type of binary phase diagram to understand and interpret is the type that is characterized by the copper–nickel system (Figure 9.3a). Temperature is plotted along the ordinate, and the abscissa represents the composition of the alloy, in weight percent of nickel. The composition ranges from 0 wt% Ni (100 wt% Cu) on the far left horizontal extreme to 100 wt% Ni (0 wt% Cu) on the right. Three different phase regions, or *fields*, appear on the diagram: an alpha (α) field, a liquid (L) field, and a two-phase $\alpha + L$ field. Each region is defined by the phase or phases that exist over the range of temperatures and compositions delineated by the phase boundary lines.

Figure 9.3: Schematic illustration of the Inverse Lever Rule applied across a two-phase ($L + \alpha$) tie-line to determine phase weight fractions W_L and W_α — from Callister & Rethwisch 10th Ed. (Fig. 9.3).

3.1 Phase Boundaries

- **Liquidus Line:** The boundary above which the alloy is completely molten **liquid** (L).
- **Solidus Line:** The boundary below which the alloy is completely solidified **solid** (α).
- **Two-Phase Region** ($L + \alpha$): The coexistence zone between liquidus and solidus.

3.2 The 2-Step Protocol for Two-Phase Quantitative Analysis

Given an overall alloy composition C_0 at temperature T inside a two-phase field:

Step 1: Determine Phase Compositions (The Tie-Line Rule)

1. Construct a horizontal isothermal line (a **tie-line**) at temperature T across the two-phase field from boundary to boundary.
2. Drop vertical lines from the tie-line intersections to the horizontal composition axis:
 - Intersection with Liquidus line gives the composition of the liquid phase: C_L .
 - Intersection with Solidus line gives the composition of the solid phase: C_α .

Step 2: Determine Phase Weight Fractions (The Inverse Lever Rule)

The weight fractions of the phases are determined by mechanical lever balance about the overall composition C_0 :

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L} \quad (\text{Weight fraction of Liquid})$$

$$W_{\alpha} = \frac{C_0 - C_L}{C_{\alpha} - C_L} \quad (\text{Weight fraction of Solid } \alpha)$$

- ☀ **The "Inverse" Lever Logic:**

- To find the fraction of the **Liquid** phase (on the left side of the tie-line), measure the length of the tie-line segment on the **opposite (right) side**: $(C_{\alpha} - C_0)$.
- To find the fraction of the **Solid** phase (on the right side), measure the segment on the **opposite (left) side**: $(C_0 - C_L)$.
- **Sum Check:** The weight fractions must always sum to unity:

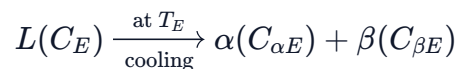
$$W_L + W_{\alpha} = \frac{C_{\alpha} - C_0}{C_{\alpha} - C_L} + \frac{C_0 - C_L}{C_{\alpha} - C_L} = \frac{C_{\alpha} - C_L}{C_{\alpha} - C_L} = 1.000 \quad (100)$$

4. Binary Eutectic Systems & Microstructures (Callister §9.10 – §9.12)

In systems where components have limited solid solubility (e.g., Lead-Tin Pb-Sn), components form a **Binary Eutectic System**.

4.1 The Eutectic Invariant Reaction

At the **eutectic point** (T_E, C_E) , a liquid phase of unique eutectic composition solidifies isothermally into two distinctly different solid phases simultaneously:



- For the Pb-Sn system: $T_E = 183^{\circ}\text{C}$, $C_E = 61.9 \text{ wt.}$
- The resulting solid microstructure consists of alternating microscopic plates (lamellae) of α and β , termed the **lamellar eutectic structure**.

4.2 Hypoeutectic vs. Hypereutectic Solidification

- **Hypoeutectic Alloys** ($C_0 < C_E$):
 - Upon cooling below the liquidus, solid α nucleates first: called **primary (proeutectic) α** .
 - As primary α grows, the remaining liquid is enriched in solute until it reaches the eutectic composition C_E at T_E .
 - At T_E , all remaining liquid transforms into the lamellar eutectic mixture $(\alpha + \beta)$.
 - *Final Microstructure*
 - : Large primary α grains surrounded by lamellar eutectic matrix.
- **Hypereutectic Alloys** ($C_0 > C_E$):
 - Forms **primary (proeutectic) β** plus lamellar eutectic mixture $(\alpha + \beta)$.

5. The Iron-Carbon (Fe — Fe₃C) Phase Diagram (Callister §9.18 – §9.19)

The Iron-Iron Carbide system is the foundation of all steel metallurgy. The diagram extends from pure iron (0 wt) to the stoichiometric intermetallic compound **Cementite** (Fe₃C) containing 6.70 wt.

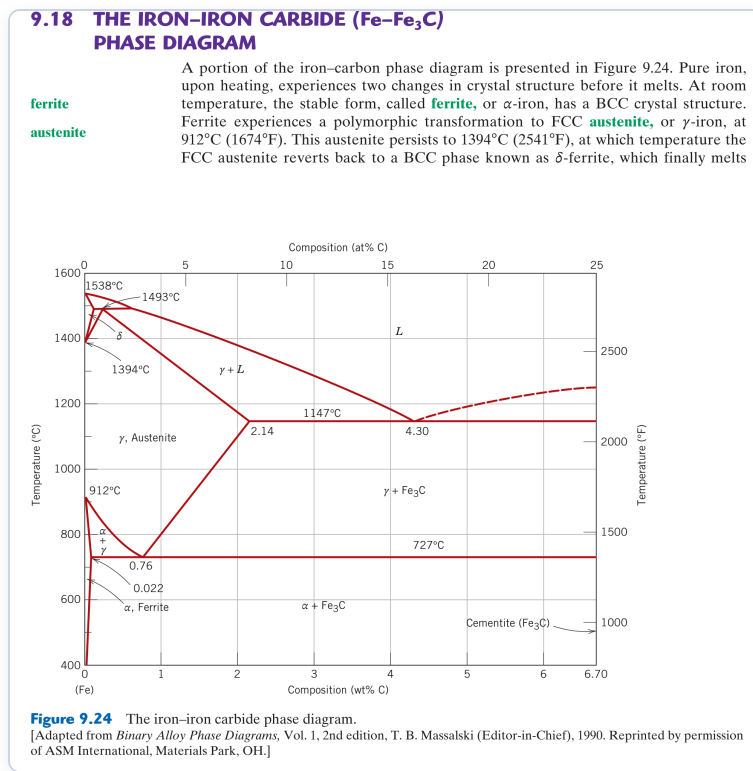


Figure 9.24: The Iron-Iron Carbide (Fe — Fe₃C) phase diagram showing α -ferrite, γ -austenite, cementite, and the eutectoid reaction at 727°C (0.76 wt) — from Callister & Rethwisch 10th Ed. (Fig. 9.24).

5.1 Phases of the Iron-Carbon System

Phase Name	Common Symbol	Crystal Structure	Carbon Solubility Limit	Mechanical Characteristics
Ferrite	α	BCC	Max 0.022 wt at 727°C (0.008 at RT)	Very soft, highly ductile, magnetic.
Austenite	γ	FCC	Max 2.14 wt at 1147°C	Ductile, easily hot-worked, non-magnetic.
δ -Ferrite	δ	BCC	Max 0.09 wt at 1493°C	Stable only at extreme temperatures (> 1394°C).
Cementite	Fe ₃ C	Orthorhombic	Fixed at 6.70 wt	Ceramic-like, extremely hard and brittle.

5.2 The Three Invariant Reactions in Fe – Fe₃C

1. Peritectic Reaction (1493°C):



2. Eutectic Reaction (1147°C):



(The resulting eutectic product is called **Ledeburite**, found in cast irons).

3. Eutectoid Reaction (727°C — THE CORE OF STEEL METALLURGY): A solid-state reaction where solid austenite (γ) decomposes into two different solid phases simultaneously:



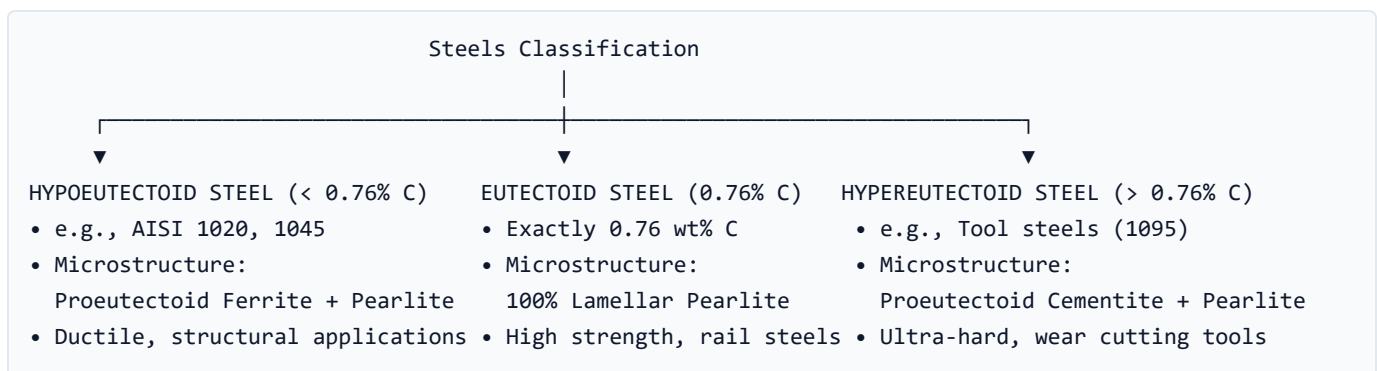
- **The Product: Pearlite:** The eutectoid product forms as alternating micro-layers (lamellae) of soft, ductile α -ferrite and hard, rigid cementite (Fe₃C). Under optical microscopy, the lamellae diffract light with a pearly sheen, hence the name **Pearlite**.

5.3 Classification of Steels by Carbon Content

Engineering iron-carbon alloys are classified into:

- **Steels:** Carbon content 0.022 wt.
- **Cast Irons:** Carbon content 2.14 wt (typically 3.0 – 4.5).

Within steels, alloys are categorized relative to the eutectoid composition (0.76 wt):



A. Hypoeutectoid Steel Microstructural Evolution ($C_0 < 0.76 \text{ wt}$)

Consider cooling a steel with $C_0 = 0.40 \text{ wt}$ (AISI 1040):

1. **Above A_3 ($T > 800^\circ\text{C}$):** Microstructure is 100 uniform austenite (γ).
2. **Between A_3 and A_1 ($727^\circ\text{C} < T < 800^\circ\text{C}$):** Austenite enters the ($\alpha + \gamma$) field. Solid α -ferrite begins to nucleate and grow preferentially along the austenite grain boundaries. Because this ferrite forms **before (pro)** the eutectoid temperature, it is called **Proeutectoid Ferrite**. As proeutectoid ferrite grows, it rejects

carbon into the surrounding austenite because ferrite can hold almost no carbon (< 0.022). The carbon content of remaining austenite rises along the A_3 boundary from 0.40 toward 0.76.

3. **At 727°C (A_1):** The remaining austenite has reached the exact eutectoid composition (0.76 wt). It transforms completely into **Pearlite**.
4. **Final Room-Temperature Microstructure:** White islands of **Proeutectoid Ferrite** surrounded by dark colonies of **Pearlite**!

B. Hypereutectoid Steel Microstructural Evolution ($C_0 > 0.76$ wt)

Consider cooling a tool steel with $C_0 = 1.10$ wt:

1. **Between A_{cm} and A_1 ($727^\circ\text{C} < T < 870^\circ\text{C}$):** Enters the $(\gamma + \text{Fe}_3\text{C})$ field. Hard cementite nucleates along austenite grain boundaries, forming a continuous network of **Proeutectoid Cementite**.
2. **At 727°C :** Remaining austenite reaches 0.76 and transforms into **Pearlite**.
3. **Final Microstructure:** A brittle boundary network of **Proeutectoid Cementite** encasing colonies of **Pearlite**.

6. Comprehensive Step-by-Step Problem Walkthroughs

6.1 Problem 1: Lever Rule Analysis of a Copper-Nickel Isomorphous Alloy

Problem Statement: A copper-nickel alloy containing $C_0 = 40$ wt is slowly cooled to a temperature of 1250°C . At 1250°C , the liquidus line composition is $C_L = 32$ wt and the solidus line composition is $C_\alpha = 45$ wt.

1. State the phases present.
2. Determine the chemical composition of each phase.
3. Calculate the weight fractions of liquid (W_L) and solid (W_α).
4. For a 5.0 kg ingot, calculate the exact mass (in kg) of solid and liquid.

Step 1: Identify Phases Present

At 1250°C , $C_0 = 40$ wt lies between $C_L = 32$ and $C_\alpha = 45$. The alloy is in the **two-phase** ($L + \alpha$) **field**. Both **Liquid** (L) and **Solid** (α) coexist.

Step 2: Determine Phase Compositions via Tie-Line

- Composition of Liquid phase: $C_L = 32$ wt (68 wt).
- Composition of Solid α phase: $C_\alpha = 45$ wt (55 wt).

Step 3: Apply the Inverse Lever Rule

Total tie-line length: $C_\alpha - C_L = 45 - 32 = 13$ wt.

- **Weight fraction of Liquid (W_L):**

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L} = \frac{45 - 40}{45 - 32} = \frac{5}{13} = 0.3846 \implies 38.5$$

- **Weight fraction of Solid α (W_α):**

$$W_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L} = \frac{40 - 32}{45 - 32} = \frac{8}{13} = 0.6154 \implies 61.5$$

- Check: $W_L + W_\alpha = 0.3846 + 0.6154 = 1.0000$ (100).

Step 4: Calculate Component Masses in a 5.0 kg Ingot

- Mass of Liquid: $m_L = W_L \times m_{\text{total}} = 0.3846 \times 5.0 \text{ kg} = 1.923 \text{ kg}$.
- Mass of Solid: $m_\alpha = W_\alpha \times m_{\text{total}} = 0.6154 \times 5.0 \text{ kg} = 3.077 \text{ kg}$.

6.2 Problem 2: Complete Quantitative Microstructural Analysis of a Hypoeutectoid Steel

Problem Statement: A hypoeutectoid plain carbon steel containing $C_0 = 0.35 \text{ wt}$ is slowly cooled to a temperature just below the eutectoid temperature ($727^\circ\text{C} - \epsilon$, say 725°C). Given:

- Maximum solubility of carbon in ferrite at 727°C : $C_\alpha = 0.022 \text{ wt}$.
- Eutectoid composition: $C_{\text{eutectoid}} = 0.76 \text{ wt}$.
- Cementite composition: $C_{\text{cem}} = 6.70 \text{ wt}$.

Calculate:

1. The weight fraction of **proeutectoid ferrite** ($W_{\alpha'}$).
2. The weight fraction of **pearlite** (W_P).
3. The **total weight fraction of ferrite** ($W_{\alpha,\text{total}}$).
4. The **total weight fraction of cementite** ($W_{\text{cem},\text{total}}$).

Step 1: Calculate Proeutectoid Ferrite ($W_{\alpha'}$) & Pearlite (W_P)

Proeutectoid ferrite forms between A_3 and A_1 . To find its fraction, we apply the lever rule **just above the eutectoid temperature** ($727^\circ\text{C} + \epsilon$), where the two phases are proeutectoid α ($C_\alpha = 0.022$) and austenite γ ($C_\gamma = 0.76$):

- **Fraction of Proeutectoid Ferrite ($W_{\alpha'}$):**

$$W_{\alpha'} = \frac{C_\gamma - C_0}{C_\gamma - C_\alpha} = \frac{0.76 - 0.35}{0.76 - 0.022} = \frac{0.410}{0.738} = 0.5556 \implies 55.6$$

- **Fraction of Austenite (γ) Just Above 727°C :**

$$W_\gamma = \frac{C_0 - C_\alpha}{C_\gamma - C_\alpha} = \frac{0.35 - 0.022}{0.76 - 0.022} = \frac{0.328}{0.738} = 0.4444 \implies 44.4$$

- **Fraction of Pearlite (W_P):** At 727°C , all remaining austenite (W_γ) transforms into Pearlite:

$$W_P = W_\gamma = 0.4444 \implies 44.4$$

- *Microstructure Check*

$$: W_{\alpha'} + W_P = 0.5556 + 0.4444 = 1.0000 \text{ (100)}.$$

Step 2: Calculate TOTAL Ferrite ($W_{\alpha,\text{total}}$) & TOTAL Cementite ($W_{\text{cem},\text{total}}$)

Just below 727°C , the steel consists fundamentally of two equilibrium phases: Ferrite (α , 0.022) and Cementite (Fe_3C , 6.70). We apply the lever rule across the **entire bottom tie-line from 0.022 to 6.70**:

- **Total Ferrite ($W_{\alpha,\text{total}}$):**

$$W_{\alpha,\text{total}} = \frac{C_{\text{cem}} - C_0}{C_{\text{cem}} - C_\alpha} = \frac{6.70 - 0.35}{6.70 - 0.022} = \frac{6.350}{6.678} = 0.9509 \Rightarrow 95.1$$

- **Total Cementite ($W_{\text{cem},\text{total}}$):**

$$W_{\text{cem},\text{total}} = \frac{C_0 - C_\alpha}{C_{\text{cem}} - C_\alpha} = \frac{0.35 - 0.022}{6.70 - 0.022} = \frac{0.328}{6.678} = 0.0491 \Rightarrow 4.91$$

- *Phase Check*

$$: W_{\alpha,\text{total}} + W_{\text{cem},\text{total}} = 0.9509 + 0.0491 = 1.0000 (100).$$

Step 3: Breakdown of Eutectoid Ferrite Inside Pearlite

Notice that total ferrite (95.1) is greater than proeutectoid ferrite (55.6). The difference is the **eutectoid ferrite residing within the pearlite lamellae**:

$$W_{\alpha,\text{eutectoid}} = W_{\alpha,\text{total}} - W_{\alpha'} = 0.9509 - 0.5556 = 0.3953 \Rightarrow 39.5$$

Pearlite itself consists of: 39.5 ferrite + 4.91 cementite = 44.4 pearlite!

7. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Proeutectoid Ferrite vs. Total Ferrite:** This is the single most common failure point on Concordia materials midterms:
 - If asked for **Proeutectoid Ferrite ($W_{\alpha'}$)**, tie-line bounds are 0.022 and 0.76 (using C_γ at the eutectoid).
 - If asked for **Total Ferrite ($W_{\alpha,\text{total}}$)**, tie-line bounds are 0.022 and 6.70 (using C_{cem}). Substituting 6.70 when calculating proeutectoid ferrite will produce an answer of 95 instead of 55.6!
- **⚠ Trap 2: Pearlite is a Microstructure, NOT a Phase:** Pearlite is NOT a phase! It is a **two-phase mixture of Ferrite (α) and Cementite (Fe_3C)**. If an exam question asks: "What phases are present at room temperature?", writing "Pearlite" is technically incorrect; the correct phases are **α -ferrite and cementite**.
- **⚠ Trap 3: Inverting the Lever Rule Numerator:** Remember: to find the phase on the left side of the tie-line (W_L or W_α), take the segment on the **right side**: ($C_{\text{right}} - C_0$). Always verify that your calculated fraction makes physical sense (if C_0 is close to the solidus, W_α should be > 50).
- **⚠ Trap 4: Cementite Composition is Fixed at 6.70 wt:** Cementite is a line compound (Fe_3C). Its carbon composition does not change with temperature; it is **always 6.70 wt**.